ATTENDEASE · SOFTWARE ENGINEERING MINI PROJECT

Unit Test Report

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Version	1.0
Date	10 November 2026
Build under test	`main` @ `92bcd23` (merge of PR #113, 1 Nov 2026)
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Results of the automated unit and integration test suite (320 tests).

1. Summary

320	320	0	34.2s
TESTS EXECUTED	PASSED	FAILED / ERRORS / SKIPPED	TOTAL TEST TIME

The complete automated suite of the AttendEase backend was executed against the final build on `main`. All **320** tests passed, with no failures, errors or skipped tests. Every test-plan case that can be automated at unit/integration level (TC-01 to TC-11 and TC-13) has at least one passing automated test, and **48 of 52** SRS requirements (FR, NFR, SEC) are exercised by at least one test. Of the other four, NFR-006 is verified by coverage measurement (TC-12), NFR-005 is structural, and NFR-008 and SEC-005 are deployment requirements (section 5).

Verdict: PASS Exit criteria of Test Plan §5.4 met: all planned tests executed, no open critical or high-severity defects.

2. Scope and environment

Unit and integration tests cover the service layer (business rules) and the REST API layer (status codes, role checks, payloads) of the five Django apps. API tests issue real HTTP requests through DRF's test client, most with real JWT logins.

Item	Detail
Operating system	Linux x86-64 (2 vCPU)
Python	3.13.17 (local run) · 3.11 in GitHub Actions CI
Framework	Django 4.2.30, Django REST Framework 3.17.2, SimpleJWT 5.5.1
Database	SQLite (Django test database, created and destroyed per run)
Test runner	Django `manage.py test` (unittest) with unittest-xml-reporting for per-test results
Coverage tool	coverage.py 7.17.0, branch measurement enabled
Static checks	Ruff 0.17.1: all checks passed · `manage.py check`: no issues
Command	`coverage run --branch --source=users,academic,attendance,corrections,analytics manage.py test`
Executed	10 November 2026, 06:32 UTC

3. Results by module

App	Area	Tests	Passed	Failed	Time (s)
users	User access (Role 1)	15	15	0	2.88
academic	Academic setup (Role 2)	66	66	0	22.98
attendance	Attendance recording & audit (Roles 2–3)	113	113	0	2.67
corrections	Corrections (Role 3)	94	94	0	0.99
analytics	Analytics & reports (Role 4)	32	32	0	4.66
Total		**320**	**320**	**0**	**34.18**

Results by test class

Module	Test class	Requirements	Tests	Passed	Time (s)
users.test_password_reset	PasswordResetTests	FR-034	3	3	0.66
users.tests	LoginTests	FR-003, SEC-002	2	2	0.72
users.tests	LogoutTests	FR-005	3	3	0.58
users.tests	RegistrationTests	FR-001, FR-002, SEC-001, SEC-003	4	4	0.33
users.tests	RoleAccessTests	FR-004, SEC-003	3	3	0.59
academic.tests	CourseTests	FR-004, FR-006, FR-035, SEC-003	15	15	4.92
academic.tests	CsvEnrollmentTests	FR-010	7	7	3.91
academic.tests	DeactivationTests	FR-012	7	7	2.32
academic.tests	DuplicateSectionCodeTests	FR-011	5	5	1.57
academic.tests	EnrollmentTests	FR-008	9	9	2.81
academic.tests	FacultyAssignmentTests	FR-013	8	8	2.69
academic.tests	SectionTests	FR-007	7	7	2.22
academic.tests	TimetableTests	FR-009	8	8	2.53
attendance.test_audit	AttendanceAuditTests	NFR-003, SEC-004, SEC-006	20	20	0.22
attendance.test_dashboard	StudentDashboardTests	FR-020, FR-021, FR-022, FR-023, NFR-007, SEC-004	9	9	0.06
attendance.test_edit_window	AttendanceEditWindowTests	FR-038	22	22	0.18
attendance.test_performance	ConcurrentLoadTests	NFR-001, NFR-002	3	3	1.38
attendance.test_performance	RosterPerformanceTests	NFR-002	3	3	0.29
attendance.test_sessions	CloseSessionTests	FR-018	5	5	0.04
attendance.test_sessions	FutureDateTests	NFR-004	5	5	0.02
attendance.test_sessions	InactiveSectionTests	FR-012	1	1	0.01

attendance.test_sessions	MarkAllPresentTests	FR-014, FR-016, FR-017	5	5	0.13
attendance.test_sessions	MarkStudentTests	FR-016	7	7	0.04
attendance.test_sessions	RosterTests	FR-015, SEC-004	5	5	0.04
attendance.test_sessions	SessionTimestampTests	FR-019	4	4	0.03
attendance.test_sessions	StartSessionTests	FR-014	9	9	0.14
attendance.tests	AttendanceSubmissionTests	FR-037, SEC-003	15	15	0.08
corrections.test_approval	CorrectionApprovalTests	FR-027, SEC-003, SEC-006	18	18	0.25
corrections.test_notifications	CorrectionNotificationTests	FR-025	16	16	0.08
corrections.test_queue	PendingCorrectionQueueTests	FR-026	19	19	0.26
corrections.test_rejection	CorrectionRejectionTests	FR-028, SEC-003, SEC-006	21	21	0.21
corrections.tests	CorrectionSubmissionTests	FR-024, SEC-004	20	20	0.18
analytics.test_exports	ReportExportApiTests	FR-033, FR-036	6	6	0.11
analytics.tests	AttendancePercentageTests	FR-029	3	3	0.49
analytics.tests	AttendanceShortageTests	FR-029, FR-030	6	6	0.98
analytics.tests	DefaulterReportApiTests	FR-029, FR-030, FR-035, FR-031, FR-036, SEC-004	9	9	2.33
analytics.tests	SectionAttendanceSummaryTests	FR-032, FR-036, SEC-004	8	8	0.76

4. Test Plan cases covered by automated tests

TC	Description	Automated test(s)	Result
TC-01	Register a new student account	`RegistrationTests.test_tc01_register_student_with_hashed_password`	PASS
TC-02	Lockout after repeated failed logins	`LoginTests.test_tc02_failed_logins_lock_account_for_cooldown`	PASS
TC-03	Admin assigns faculty; faculty sees the section	`FacultyAssignmentTests.test_tc03_admin_assigns_faculty_who_then_sees_section`	PASS
TC-04	Start session and bulk-mark all present	`MarkAllPresentTests.test_tc04_start_session_then_bulk_mark_all_present`	PASS
TC-05	Duplicate attendance record rejected (409)	`AttendanceSubmissionTests.test_tc05_repeated_submission_returns_409_without_overwrite`	PASS
TC-06	Student attendance dashboard	`StudentDashboardTests.test_tc06_course_percentage_counts_only_closed_sessions` `StudentDashboardTests.test_tc06_session_history_is_closed_and_student_scoped`	PASS

TC-07	Student raises a correction; faculty emailed	`CorrectionNotificationTests.test_tc07_success_sends_one_email_only_to_assigned_faculty` `CorrectionSubmissionTests.test_tc07_valid_request_is_pending_and_leaves_attendance_unchanged`	PASS
TC-08	Faculty approves a correction; audit fields set	`CorrectionApprovalTests.test_tc08_approval_updates_attendance_and_records_server_audit_fields`	PASS
TC-09	Student calling an admin-only endpoint gets 403	`CourseTests.test_tc09_student_calling_real_admin_endpoint_is_denied` `RoleAccessTests.test_tc09_student_cannot_call_admin_only_endpoint`	PASS
TC-10	Defaulter report for a section	`DefaulterReportApiTests.test_report_lists_only_below_threshold_with_required_columns`	PASS

TC-11	50 concurrent users; 80-student roster	`ConcurrentLoadTests.test_nfr001_mixed_api_requests_under_concurrent_load`
		`ConcurrentLoadTests.test_tc11_50_concurrent_roster_requests_within_2_seconds`
		`RosterPerformanceTests.test_nfr002_marking_actions_for_80_students_stay_within_3_seconds`
		`RosterPerformanceTests.test_nfr002_roster_for_80_students_loads_within_3_seconds`
		`RosterPerformanceTests.test_nfr002_roster_query_count_does_not_grow_with_class_size`
			PASS

TC-12	Service-layer coverage ≥ 70%	Measured by coverage.py (see Coverage Reports)	PASS

TC-13	Student cannot access another student's record	`AttendanceAuditTests.test_history_is_scoped_to_student_assigned_faculty_and_admin`
		`StudentDashboardTests.test_student_only_sees_enrolled_courses`
		`RosterTests.test_fr015_roster_visible_to_assigned_faculty_and_admin_only`
		`CorrectionSubmissionTests.test_another_students_record_is_not_disclosed`
			PASS

5. Requirement coverage

Requirement coverage (Test Plan §14): **48/52 = 92%** of FR/NFR/SEC IDs have at least one passing test.

ID	Requirement	Tests	Status
FR-001	Register with name, email and role	3	Covered
FR-002	Email uniqueness at registration	1	Covered
FR-003	Log in with email and password	2	Covered
FR-004	Role-based access control on every protected endpoint	4	Covered
FR-005	Log out, invalidating the token	3	Covered
FR-034	Reset a forgotten password via email	3	Covered
FR-020	Attendance percentage per enrolled course	5	Covered
FR-021	Session-wise attendance history	5	Covered
FR-022	Flag courses below the shortage threshold	1	Covered

FR-023	Filter history by date range	2	Covered
FR-006	Create and edit courses	8	Covered
FR-007	Create and edit sections	7	Covered
FR-008	Enroll students into a section	9	Covered
FR-009	Weekly timetable per section	8	Covered
FR-010	Bulk enrollment via CSV	7	Covered
FR-011	No duplicate section codes within a course	5	Covered
FR-012	Deactivate a course or section	8	Covered
FR-013	Assign a faculty member to a section	8	Covered
FR-014	Start a session for a scheduled period	10	Covered
FR-015	Roster for the active session	5	Covered
FR-016	Mark each student present or absent	8	Covered
FR-017	Bulk-mark all present in one action	5	Covered
FR-018	Close / finalize a session	5	Covered
FR-019	Timestamp session creation and closure	4	Covered
FR-037	No duplicate records per student and session	15	Covered
FR-038	Edit within a configurable window after closure	22	Covered
FR-024	Student raises a correction request	20	Covered
FR-025	Email the faculty member on a new request	16	Covered
FR-026	Pending-request queue for faculty	19	Covered
FR-027	Approve a request, updating the record	18	Covered
FR-028	Reject a request with a reason	21	Covered
FR-029	Compute attendance percentage per course	10	Covered
FR-030	Identify students below the threshold	15	Covered
FR-035	Per-course threshold set by an administrator	7	Covered
FR-031	Defaulter report for a section	9	Covered
FR-032	Section-wise summary report	8	Covered
FR-033	Export reports as CSV / PDF	6	Covered
FR-036	Filter reports by date range and section	3	Covered
NFR-001	95% of requests within 2 s at 50 concurrent users	3	Covered
NFR-002	Roster for 80 students within 3 s	4	Covered
NFR-003	No permanent deletion; auditable corrections	20	Covered
NFR-004	No attendance for future dates	5	Covered

NFR-005	Each module independently testable Structural: each app has its own test modules (table 3) and runs in isolation, e.g. `manage.py test corrections`.	–	Not unit-tested
NFR-006	At least 70% service-layer coverage Measured by coverage, see Code and Branch Coverage Reports (TC-12).	–	Not unit-tested
NFR-007	Usable on desktop and tablet widths	1	Covered
NFR-008	Deployable with one docker-compose command Deployment requirement (`docker-compose up`); outside the scope of the automated suite.	–	Not unit-tested
SEC-001	Salted one-way password hashing	1	Covered
SEC-002	Lock out / throttle repeated failed logins	1	Covered
SEC-003	Server-side RBAC on every write endpoint	8	Covered
SEC-004	Queries scoped to the user's own sections	6	Covered
SEC-005	All traffic over HTTPS Deployment requirement (TLS at the web server / proxy); no automated test. See open items.	–	Not unit-tested
SEC-006	Audit trail (acted_by, acted_at) on decisions	59	Covered

6. Defects and observations

No defects were found in this run. Defects raised during development were fixed in their feature branches before merging, so the CI pass rate on `main` is 100%. Slowest tests (the performance tests deliberately create load):

Test	Time (s)
`ConcurrentLoadTests.test_tc11_50_concurrent_roster_requests_within_2_seconds`	0.72
`CsvEnrollmentTests.test_fr010_only_admins_can_upload`	0.57
`LoginTests.test_tc02_failed_logins_lock_account_for_cooldown`	0.57
`CsvEnrollmentTests.test_fr010_missing_email_header_or_empty_file_is_rejected`	0.56
`CsvEnrollmentTests.test_fr010_missing_file_and_non_utf8_are_rejected`	0.56
`CsvEnrollmentTests.test_fr010_inactive_section_rejects_upload`	0.56
`CsvEnrollmentTests.test_fr010_admin_enrolls_students_from_csv`	0.55
`CsvEnrollmentTests.test_fr010_invalid_rows_reject_whole_file`	0.55

Appendix A. Full test list

Test names are shown without the `test_` prefix. Requirements are derived from the test name and class.

Test	Requirements · TC	Time (s)	Result

users.test_password_reset.PasswordResetTests			
fr034_confirm_changes_password_once_and_revokes_jwt	FR-034	0.413	Passed
fr034_rejects_bad_token_and_weak_password	FR-034	0.163	Passed
fr034_request_does_not_reveal_unknown_email	FR-034	0.087	Passed
users.tests.LoginTests			
fr003_login_issues_jwt_pair	FR-003	0.158	Passed
tc02_failed_logins_lock_account_for_cooldown	FR-003, SEC-002 · TC-02	0.566	Passed
users.tests.LogoutTests			
fr005_logout_revokes_access_and_refresh_tokens	FR-005	0.250	Passed
logout_requires_authentication	FR-005	0.163	Passed
refresh_works_before_logout	FR-005	0.163	Passed
users.tests.RegistrationTests			
fr002_rejects_duplicate_email_ignoring_case	FR-002	0.082	Passed
public_registration_cannot_create_admin	FR-001, SEC-003	0.003	Passed
registration_stores_optional_usn	FR-001	0.083	Passed
tc01_register_student_with_hashed_password	FR-001, SEC-001 · TC-01	0.158	Passed
users.tests.RoleAccessTests			
admin_can_call_admin_only_endpoint	FR-004, SEC-003	0.156	Passed
protected_me_endpoint_requires_jwt	FR-004, SEC-003	0.244	Passed
tc09_student_cannot_call_admin_only_endpoint	FR-004, SEC-003 · TC-09	0.195	Passed
academic.tests.CourseTests			
course_endpoints_require_authentication	FR-006	0.351	Passed
fr006_admin_creates_course	FR-006	0.333	Passed
fr006_admin_edits_course	FR-006	0.333	Passed
fr006_any_role_lists_courses	FR-006	0.318	Passed
fr006_duplicate_course_code_is_rejected	FR-006	0.316	Passed
fr006_invalid_course_payload_is_rejected	FR-006	0.315	Passed
fr006_non_admins_cannot_create_or_edit_courses	FR-006	0.322	Passed
fr006_unknown_course_returns_404	FR-006	0.317	Passed
fr035_admin_configures_threshold_on_course_creation	FR-035	0.315	Passed
fr035_admin_updates_course_threshold	FR-035	0.314	Passed
fr035_threshold_accepts_percentage_boundaries	FR-035	0.322	Passed

fr035_threshold_cannot_be_configured_by_non_admins	FR-035	0.315	Passed
fr035_threshold_defaults_to_existing_value_when_omitted	FR-035	0.316	Passed
fr035_threshold_rejects_out_of_range_and_excess_precision	FR-035	0.322	Passed
tc09_student_calling_real_admin_endpoint_is_denied	FR-004, SEC-003 · TC-09	0.408	Passed
academic.tests.CsvEnrollmentTests			
fr010_admin_enrolls_students_from_csv	FR-010	0.550	Passed
fr010_already_enrolled_students_are_skipped	FR-010	0.545	Passed
fr010_inactive_section_rejects_upload	FR-010	0.562	Passed
fr010_invalid_rows_reject_whole_file	FR-010	0.548	Passed
fr010_missing_email_header_or_empty_file_is_rejected	FR-010	0.565	Passed
fr010_missing_file_and_non_utf8_are_rejected	FR-010	0.564	Passed
fr010_only_admins_can_upload	FR-010	0.571	Passed
academic.tests.DeactivationTests			
fr012_admin_deactivates_and_reactivates_section	FR-012	0.334	Passed
fr012_admin_deactivates_course	FR-012	0.315	Passed
fr012_data_is_kept_when_deactivated	FR-012	0.310	Passed
fr012_inactive_course_hidden_with_its_sections	FR-012	0.318	Passed
fr012_inactive_items_hidden_from_non_admins	FR-012	0.328	Passed
fr012_no_new_enrollments_periods_or_sections_when_inactive	FR-012	0.403	Passed
fr012_only_admins_can_deactivate	FR-012	0.316	Passed
academic.tests.DuplicateSectionCodeTests			
fr011_database_enforces_unique_section_code	FR-011	0.316	Passed
fr011_duplicate_section_code_in_same_course_is_rejected	FR-011	0.314	Passed
fr011_renaming_section_to_its_own_code_is_allowed	FR-011	0.314	Passed
fr011_renaming_to_sibling_code_is_rejected	FR-011	0.314	Passed
fr011_same_section_code_allowed_in_different_course	FR-011	0.315	Passed
academic.tests.EnrollmentTests			
fr008_admin_enrolls_student	FR-008	0.315	Passed
fr008_admin_unenrolls_student	FR-008	0.315	Passed
fr008_duplicate_enrollment_is_rejected	FR-008	0.314	Passed
fr008_inactive_student_cannot_be_enrolled	FR-008	0.314	Passed
fr008_non_admin_cannot_enroll	FR-008	0.312	Passed
fr008_only_student_accounts_can_be_enrolled	FR-008	0.313	Passed
fr008_roster_visible_to_admin_and_assigned_faculty_only	FR-008	0.315	Passed
fr008_service_rejects_non_student	FR-008	0.305	Passed

fr008_unknown_section_returns_404	FR-008	0.311	Passed
academic.tests.FacultyAssignmentTests			
admins_have_no_personal_section_list	FR-013	0.308	Passed
enrolled_student_sees_section_in_my_sections	FR-013	0.313	Passed
fr013_non_admin_cannot_assign_faculty	FR-013	0.321	Passed
fr013_only_faculty_accounts_can_be_assigned	FR-013	0.315	Passed
fr013_reassignment_moves_section_between_faculty	FR-013	0.320	Passed
fr013_service_rejects_non_faculty	FR-013	0.307	Passed
fr013_unknown_section_is_rejected	FR-013	0.330	Passed
tc03_admin_assigns_faculty_who_then_sees_section	FR-013 · TC-03	0.480	Passed
academic.tests.SectionTests			
fr007_admin_creates_section_without_faculty	FR-007	0.314	Passed
fr007_admin_edits_section_code	FR-007	0.315	Passed
fr007_course_list_includes_sections	FR-007	0.319	Passed
fr007_lists_sections_for_course	FR-007	0.321	Passed
fr007_non_admins_cannot_create_or_edit_sections	FR-007	0.323	Passed
fr007_section_code_is_required	FR-007	0.315	Passed
fr007_unknown_course_or_section_returns_404	FR-007	0.313	Passed
academic.tests.TimetableTests			
fr009_admin_defines_weekly_timetable	FR-009	0.325	Passed
fr009_admin_removes_period	FR-009	0.312	Passed
fr009_end_time_must_follow_start_time	FR-009	0.312	Passed
fr009_invalid_day_is_rejected	FR-009	0.316	Passed
fr009_non_admin_cannot_change_timetable	FR-009	0.313	Passed
fr009_other_sections_can_use_same_period	FR-009	0.319	Passed
fr009_overlapping_period_in_same_section_is_rejected	FR-009	0.322	Passed
fr009_service_detects_overlap	FR-009	0.310	Passed
attendance.test_audit.AttendanceAuditTests			
approval_preserves_old_value_and_links_decision	NFR-003, SEC-006	0.008	Passed
audit_model_and_queryset_changes_are_blocked	NFR-003, SEC-006	0.005	Passed
batch_submission_records_each_initial_mark	NFR-003, SEC-006	0.004	Passed
bulk_mark_records_created_and_changed_values	NFR-003, SEC-006	0.009	Passed

cascading_deletions_cannot_erase_attendance	NFR-003, SEC-006	0.014	Passed
existing_records_receive_honest_unknown_actor_baseline	NFR-003, SEC-006	0.002	Passed
failed_audit_insert_rolls_back_approval_and_attendance	NFR-003, SEC-006	0.007	Passed
failed_audit_insert_rolls_back_mark_creation	NFR-003, SEC-006	0.004	Passed
failed_audit_insert_rolls_back_rejection	NFR-003, SEC-006	0.006	Passed
failed_bulk_audit_write_rolls_back_bulk_attendance	NFR-003, SEC-006	0.004	Passed
failed_edit_audit_write_preserves_original_mark	NFR-003, SEC-006	0.006	Passed
history_is_paginated_and_read_only	NFR-003, SEC-006	0.082	Passed
history_is_scoped_to_student_assigned_faculty_and_admin	NFR-003, SEC-004, SEC-006 · TC-13	0.017	Passed
initial_mark_and_changes_form_ordered_history	NFR-003, SEC-006	0.009	Passed
legacy_record_without_audit_cannot_be_deleted	NFR-003, SEC-006	0.003	Passed
no_op_marking_and_bulk_mark_do_not_invent_changes	NFR-003, SEC-006	0.010	Passed
record_instance_and_queryset_deletion_are_blocked	NFR-003, SEC-006	0.006	Passed
rejection_logs_decision_even_without_mark_change	NFR-003, SEC-006	0.009	Passed
timed_edit_preserves_original_value_and_actor	NFR-003, SEC-006	0.007	Passed
upsert_cannot_replace_audit_history	NFR-003, SEC-006	0.004	Passed
attendance.test_dashboard.StudentDashboardTests			
dashboard_is_student_only	FR-020, FR-021	0.005	Passed
fr022_flags_courses_below_their_own_threshold	FR-022	0.011	Passed
fr023_history_filters_inclusive_date_range	FR-023	0.007	Passed
fr023_rejects_bad_or_reversed_dates	FR-023	0.003	Passed
history_requires_student_authentication	FR-020, FR-021	0.004	Passed
nfr007_student_portal_is_responsive_and_links_to_protected_apis	NFR-007	0.010	Passed

student_only_sees_enrolled_courses	FR-020, FR-021, SEC-004 · TC-13	0.007	Passed
tc06_course_percentage_counts_only_closed_sessions	FR-020, FR-021 · TC-06	0.006	Passed
tc06_session_history_is_closed_and_student_scoped	FR-020, FR-021 · TC-06	0.005	Passed
attendance.test_edit_window.AttendanceEditWindowTests			
assigned_faculty_can_edit_without_reopening_or_duplicate	FR-038	0.008	Passed
caller_cannot_extend_window_or_change_record_identity	FR-038	0.003	Passed
can_change_back_to_absent_within_window	FR-038	0.006	Passed
closed_session_put_remains_blocked	FR-038	0.003	Passed
current_assignment_controls_access	FR-038	0.005	Passed
deadline_boundary_is_exclusive	FR-038	0.010	Passed
edit_does_not_change_another_students_record	FR-038	0.005	Passed
exact_closure_time_is_allowed	FR-038	0.005	Passed
future_closure_timestamp_is_rejected	FR-038	0.004	Passed
future_session_date_is_rejected	FR-038	0.003	Passed
inactive_faculty_cannot_edit	FR-038	0.003	Passed
invalid_or_missing_status_is_rejected	FR-038	0.004	Passed
longer_configured_window_allows_older_closed_session	FR-038	0.004	Passed
missing_closure_timestamp_is_rejected	FR-038	0.004	Passed
open_session_uses_normal_put_instead	FR-038	0.006	Passed
real_faculty_jwt_can_edit	FR-038	0.087	Passed
service_checks_permissions_and_status_independently	FR-038	0.002	Passed
shorter_configured_window_is_enforced	FR-038	0.003	Passed
student_admin_and_anonymous_cannot_edit	FR-038	0.006	Passed
unknown_session_and_missing_record_return_404_without_creating_records	FR-038	0.005	Passed
wrong_faculty_cannot_edit	FR-038	0.003	Passed
zero_disables_direct_edits	FR-038	0.003	Passed
attendance.test_performance.ConcurrentLoadTests			
nfr001_mixed_api_requests_under_concurrent_load	NFR-001	0.481	Passed
percentile_helper	NFR-001	0.177	Passed
tc11_50_concurrent_roster_requests_within_2_seconds	NFR-001, NFR-002 · TC-11	0.724	Passed
attendance.test_performance.RosterPerformanceTests			
nfr002_marking_actions_for_80_students_stay_within_3_seconds	NFR-002	0.112	Passed

nfr002_roster_for_80_students_loads_within_3_seconds	NFR-002	0.091	Passed
nfr002_roster_query_count_does_not_grow_with_class_size	NFR-002	0.087	Passed
attendance.test_sessions.CloseSessionTests			
fr018_closed_session_counts_towards_attendance_percentage	FR-018	0.015	Passed
fr018_closed_session_rejects_further_marking	FR-018	0.010	Passed
fr018_closing_twice_returns_409	FR-018	0.006	Passed
fr018_faculty_closes_session	FR-018	0.010	Passed
fr018_only_assigned_faculty_can_close	FR-018	0.003	Passed
attendance.test_sessions.FutureDateTests			
nfr004_future_session_date_is_rejected	NFR-004	0.004	Passed
nfr004_missing_date_defaults_to_today	NFR-004	0.005	Passed
nfr004_records_cannot_be_submitted_for_a_future_session	NFR-004	0.004	Passed
nfr004_service_rejects_future_date	NFR-004	0.001	Passed
nfr004_today_and_past_dates_are_allowed	NFR-004	0.008	Passed
attendance.test_sessions.InactiveSectionTests			
fr012_no_session_for_inactive_section_or_course	FR-012	0.007	Passed
attendance.test_sessions.MarkAllPresentTests			
fr017_closed_session_and_other_faculty_are_rejected	FR-017	0.006	Passed
fr017_individual_student_can_then_be_marked_absent	FR-017	0.011	Passed
fr017_is_safe_to_repeat	FR-017	0.012	Passed
fr017_overrides_earlier_absent_marks_without_duplicates	FR-017	0.008	Passed
tc04_start_session_then_bulk_mark_all_present	FR-014, FR-016, FR-017 · TC-04	0.093	Passed
attendance.test_sessions.MarkStudentTests			
fr016_closed_session_cannot_be_marked	FR-016	0.004	Passed
fr016_faculty_marks_each_student_present_or_absent	FR-016	0.009	Passed
fr016_invalid_status_is_rejected	FR-016	0.003	Passed
fr016_only_assigned_faculty_can_mark	FR-016	0.005	Passed
fr016_roster_reflects_marks	FR-016	0.008	Passed
fr016_toggling_updates_the_same_record	FR-016	0.009	Passed
fr016_unenrolled_student_returns_404	FR-016	0.005	Passed
attendance.test_sessions.RosterTests			
fr015_roster_lists_enrolled_students_with_status	FR-015	0.007	Passed
fr015_roster_uses_constant_number_of_queries	FR-015	0.012	Passed
fr015_roster_visible_to_assigned_faculty_and_admin_only	FR-015, SEC-004 · TC-13	0.013	Passed

fr015_status_is_per_session	FR-015	0.005	Passed
fr015_unenrolled_and_inactive_students_are_excluded	FR-015	0.007	Passed
attendance.test_sessions.SessionTimestampTests			
fr019_closure_is_timestamped	FR-019	0.008	Passed
fr019_creation_is_timestamped	FR-019	0.006	Passed
fr019_failed_second_close_keeps_original_timestamp	FR-019	0.009	Passed
fr019_timestamps_cannot_be_set_by_the_client	FR-019	0.006	Passed
attendance.test_sessions.StartSessionTests			
fr014_date_must_match_period_weekday	FR-014	0.005	Passed
fr014_faculty_starts_session_for_scheduled_period	FR-014	0.008	Passed
fr014_only_the_assigned_faculty_can_start	FR-014	0.003	Passed
fr014_same_period_on_another_week_is_allowed	FR-014	0.010	Passed
fr014_second_session_for_same_period_and_date_returns_409	FR-014	0.008	Passed
fr014_session_list_and_detail_are_scoped	FR-014	0.013	Passed
fr014_students_and_admins_cannot_start	FR-014	0.003	Passed
fr014_unknown_period_is_rejected	FR-014	0.003	Passed
fr014_works_with_real_jwt_login	FR-014	0.089	Passed
attendance.tests.AttendanceSubmissionTests			
anonymous_request_requires_authentication	FR-037	0.006	Passed
bad_payloads_are_rejected_without_writes	FR-037	0.006	Passed
closed_session_is_rejected	FR-037	0.004	Passed
database_constraint_protects_direct_writes	FR-037	0.007	Passed
duplicate_inside_batch_returns_409_and_saves_nothing	FR-037	0.004	Passed
existing_duplicate_rejects_whole_batch	FR-037	0.009	Passed
inactive_student_is_rejected	FR-037	0.004	Passed
missing_records_is_rejected	FR-037	0.002	Passed
other_faculty_cannot_mark_this_section	FR-037	0.004	Passed
same_student_can_have_records_in_different_sessions	FR-037	0.010	Passed
student_and_admin_cannot_submit_attendance	FR-037, SEC-003	0.004	Passed
tc05_repeated_submission_returns_409_without_overwrite	FR-037 · TC-05	0.009	Passed
unenrolled_student_rejects_whole_batch	FR-037	0.004	Passed
unknown_session_returns_404	FR-037	0.003	Passed
valid_batch_saves_present_and_absent	FR-037	0.006	Passed
corrections.test_approval.CorrectionApprovalTests			

already_present_record_can_be_approved_without_duplicate	FR-027, SEC-006	0.008	Passed
anonymous_requires_authentication	FR-027, SEC-006	0.006	Passed
approval_preserves_other_students_attendance	FR-027, SEC-006	0.006	Passed
approved_request_leaves_queue_and_updates_existing_analytics	FR-027, SEC-006	0.016	Passed
client_cannot_forge_attendance_or_decision_metadata	FR-027, SEC-006	0.007	Passed
current_assignment_controls_access	FR-027, SEC-006	0.010	Passed
failed_decision_save_rolls_back_attendance_change	FR-027, SEC-006	0.004	Passed
inactive_faculty_cannot_approve	FR-027, SEC-006	0.004	Passed
invalid_or_missing_action_is_rejected	FR-027, SEC-006	0.006	Passed
real_faculty_jwt_can_approve	FR-027, SEC-006	0.091	Passed
rejected_request_cannot_be_approved	FR-027, SEC-006	0.058	Passed
repeated_approval_returns_409_and_preserves_decision	FR-027, SEC-006	0.010	Passed
service_enforces_role_independently_of_view	FR-027, SEC-006	0.002	Passed
student_and_admin_cannot_approve	FR-027, SEC-003, SEC-006	0.004	Passed
tc08_approval_updates_attendance_and_records_server_audit_fields	FR-027, SEC-006 · TC-08	0.007	Passed
unassigned_section_cannot_be_approved	FR-027, SEC-006	0.005	Passed
unknown_request_returns_404	FR-027, SEC-006	0.004	Passed
wrong_faculty_cannot_approve	FR-027, SEC-006	0.005	Passed
corrections.test_notifications.CorrectionNotificationTests			
custom_backend_error_is_nonfatal_and_retried	FR-025	0.005	Passed
duplicate_request_registers_no_email	FR-025	0.005	Passed
faculty_with_no_email_does_not_receive_email	FR-025	0.004	Passed
inactive_faculty_does_not_receive_email	FR-025	0.004	Passed
invalid_reason_registers_no_email	FR-025	0.002	Passed

missing_request_is_logged_without_sending	FR-025	0.003	Passed
network_failure_is_retried_once	FR-025	0.005	Passed
non_faculty_assignee_does_not_receive_email	FR-025	0.004	Passed
permanent_smtp_failure_preserves_created_request_and_201	FR-025	0.008	Passed
successful_delivery_is_not_retried	FR-025	0.004	Passed
tc07_success_sends_one_email_only_to_assigned_faculty	FR-025 · TC-07	0.013	Passed
temporary_smtp_error_retries_once_then_succeeds	FR-025	0.005	Passed
transaction_rollback_sends_no_email	FR-025	0.004	Passed
unassigned_section_logs_failure_without_breaking_submission	FR-025	0.006	Passed
unauthorized_user_registers_no_email	FR-025	0.002	Passed
zero_messages_sent_is_retried_and_logged	FR-025	0.004	Passed
corrections.test_queue.PendingCorrectionQueueTests			
anonymous_request_requires_authentication	FR-026	0.005	Passed
approved_and_rejected_requests_are_excluded	FR-026	0.006	Passed
assigned_section_with_no_pending_requests_returns_empty_queue	FR-026	0.005	Passed
default_pagination_is_25_and_second_page_has_remaining_request	FR-026	0.025	Passed
inactive_faculty_is_rejected	FR-026	0.002	Passed
invalid_page_returns_404	FR-026	0.003	Passed
multiple_assigned_sections_are_included	FR-026	0.006	Passed
no_assigned_sections_returns_empty_queue	FR-026	0.005	Passed
only_own_pending_requests_are_returned_with_review_context	FR-026	0.006	Passed
other_faculty_gets_only_their_own_queue	FR-026	0.005	Passed
page_size_is_capped_at_100	FR-026	0.061	Passed
query_parameters_cannot_expand_access_or_include_resolved_requests	FR-026	0.007	Passed
real_faculty_jwt_can_read_queue	FR-026	0.091	Passed
reassignment_removes_old_faculty_access_immediately	FR-026	0.009	Passed
rendering_queue_avoids_per_item_database_queries	FR-026	0.009	Passed
requests_are_ordered_oldest_first_with_stable_ties	FR-026	0.008	Passed
service_rejects_non_faculty_independently_of_api	FR-026	0.001	Passed
students_and_admins_cannot_read_the_faculty_queue	FR-026	0.002	Passed
viewing_queue_does_not_change_requests_or_attendance	FR-026	0.006	Passed
corrections.test_rejection.CorrectionRejectionTests			
anonymous_requires_authentication	FR-028, SEC-006	0.007	Passed
approved_request_cannot_be_rejected	FR-028, SEC-006	0.010	Passed

caller_cannot_forge_decision_metadata_or_replace_student_reason	FR-028, SEC-006	0.006	Passed
failed_save_leaves_pending_request_and_attendance_unchanged	FR-028, SEC-006	0.004	Passed
inactive_faculty_cannot_reject	FR-028, SEC-006	0.003	Passed
maximum_reason_length_is_accepted	FR-028, SEC-006	0.005	Passed
missing_blank_null_and_oversized_reasons_are_rejected	FR-028, SEC-006	0.007	Passed
real_faculty_jwt_can_reject	FR-028, SEC-006	0.090	Passed
reassignment_moves_rejection_permission	FR-028, SEC-006	0.009	Passed
rejected_request_cannot_be_approved	FR-028, SEC-006	0.009	Passed
rejection_preserves_present_record_too	FR-028, SEC-006	0.005	Passed
rejection_removes_queue_item_without_changing_attendance_percentage	FR-028, SEC-006	0.011	Passed
rejection_saves_separate_reason_and_server_audit_fields	FR-028, SEC-006	0.006	Passed
repeated_rejection_preserves_original_decision	FR-028, SEC-006	0.009	Passed
service_enforces_role_independently_of_api	FR-028, SEC-006	0.002	Passed
service_validates_reason_independently_of_api	FR-028, SEC-006	0.002	Passed
student_can_raise_new_request_after_rejection_and_old_reason_is_preserved	FR-028, SEC-006	0.008	Passed
students_and_admin_cannot_reject	FR-028, SEC-003, SEC-006	0.004	Passed
unassigned_section_cannot_be_rejected	FR-028, SEC-006	0.005	Passed
unknown_request_returns_404	FR-028, SEC-006	0.004	Passed
wrong_faculty_cannot_reject	FR-028, SEC-006	0.005	Passed
corrections.tests.CorrectionSubmissionTests			
anonymous_request_requires_authentication	FR-024	0.005	Passed
another_students_record_is_not_disclosed	FR-024, SEC-004 · TC-13	0.003	Passed
attendance_record_with_request_is_protected_from_deletion	FR-024	0.006	Passed

blank_null_and_oversized_reasons_are_rejected	FR-024	0.006	Passed
client_cannot_override_identity_status_or_audit_fields	FR-024	0.005	Passed
database_also_blocks_two_pending_requests	FR-024	0.005	Passed
distinct_records_can_each_have_a_pending_request	FR-024	0.009	Passed
duplicate_pending_request_returns_409_and_preserves_original	FR-024	0.008	Passed
faculty_and_admin_cannot_submit_student_requests	FR-024	0.003	Passed
inactive_student_is_rejected	FR-024	0.003	Passed
invalid_record_ids_are_rejected	FR-024	0.007	Passed
maximum_length_reason_is_accepted	FR-024	0.004	Passed
missing_required_fields_are_rejected	FR-024	0.004	Passed
nonexistent_record_returns_404	FR-024	0.003	Passed
real_jwt_authentication_can_submit_request	FR-024	0.089	Passed
reason_is_trimmed	FR-024	0.005	Passed
resolved_requests_do_not_block_a_new_pending_request	FR-024	0.007	Passed
service_rejects_blank_reason_independently_of_api	FR-024	0.001	Passed
tc07_valid_request_is_pending_and_leaves_attendance_unchanged	FR-024 · TC-07	0.006	Passed
unenrolled_student_cannot_request_correction	FR-024	0.004	Passed
analytics.test_exports.ReportExportApiTests			
defaulter_csv_has_documented_columns_and_api_values	FR-033	0.018	Passed
defaulter_pdf_is_downloadable_and_contains_report_values	FR-033	0.016	Passed
empty_reports_still_export_headers_or_zero_summary	FR-033	0.023	Passed
invalid_format_dates_and_section_ids_are_rejected	FR-033	0.008	Passed
json_default_and_export_authorization_are_preserved	FR-033	0.014	Passed
summary_csv_and_pdf_respect_date_filters_and_api_values	FR-033, FR-036	0.027	Passed
analytics.tests.AttendancePercentageTests			
no_closed_sessions_returns_zero	FR-029	0.170	Passed
open_sessions_are_not_counted	FR-029	0.157	Passed
percentage_is_75_for_18_of_24_sessions	FR-029	0.167	Passed
analytics.tests.AttendanceShortageTests			
no_closed_sessions_returns_zero	FR-029, FR-030	0.160	Passed
open_sessions_are_not_counted	FR-029, FR-030	0.159	Passed
percentage_is_75_for_18_of_24_sessions	FR-029, FR-030	0.167	Passed

student_above_threshold_is_not_identified	FR-029, FR-030	0.165	Passed
student_at_threshold_is_not_identified	FR-029, FR-030	0.164	Passed
student_below_threshold_is_identified	FR-029, FR-030	0.163	Passed
analytics.tests.DefaulterReportApiTests			
authentication_and_section_authorization_are_enforced	FR-030, FR-031, SEC-004	0.261	Passed
date_filters_change_defaulter_calculations	FR-030, FR-031, FR-036	0.284	Passed
invalid_date_formats_and_reversed_ranges_are_rejected_on_reports	FR-030, FR-031	0.255	Passed
missing_malformed_and_unknown_section_ids_are_rejected	FR-030, FR-031	0.279	Passed
report_is_limited_to_students_enrolled_in_requested_section	FR-030, FR-031	0.238	Passed
report_lists_only_below_threshold_with_required_columns	FR-029, FR-030, FR-031 · TC-10	0.254	Passed
students_at_or_above_threshold_are_not_reported	FR-030, FR-031	0.260	Passed
threshold_is_taken_from_each_courses_configuration	FR-030, FR-035, FR-031	0.252	Passed
zero_threshold_does_not_classify_zero_attendance_as_below	FR-030, FR-031	0.243	Passed
analytics.tests.SectionAttendanceSummaryTests			
date_filters_apply_before_summary_calculations	FR-032, FR-036	0.126	Passed
empty_section_returns_zero_and_null_date_range	FR-032	0.086	Passed
missing_malformed_and_unknown_section_ids_are_rejected	FR-032	0.088	Passed
missing_records_count_as_absences_and_no_sessions_return_zero	FR-032	0.098	Passed
only_assigned_faculty_and_administrators_can_view_summary	FR-032, SEC-004	0.098	Passed
perfect_attendance_returns_one_hundred_percent	FR-032	0.086	Passed
report_counts_sessions_only_for_the_requested_section	FR-032	0.087	Passed
summary_averages_students_and_uses_closed_section_sessions	FR-032	0.087	Passed